

Simulated Annealing

ENEL445 Applied Engineering Optimisation

2025

Outline

- Introduction to Statistical Mechanics
- Simulated Annealing
- Applications

Statistical Mechanics

Statistical Mechanics

- A box with 100 identical coins.
- Shake it.
- Some will end up heads, some tails.
- Assume each of these different configurations is equally likely.



Statistical Mechanics

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

HHHTTT... 50 heads, 50 tails

THHTTH... 53 heads, 47 tails

HTTTTH... 90 heads, 10 tails

HHHHHH... 100 heads, 0 tails

- The system could be described by a very large number of *equally likely* **microstates**.
- What we actually measure is the **macrostates** of the system, which are *not equally likely*.

Statistical Mechanics

- **Microstates** - position of particles, velocity of particles
- **Macrostates** - pressure, temperature, volume, density
- The most likely *macrostate* that the system will find itself in is the one that corresponds to the largest number of *microstates*.
- In general it is impossible to measure which microstate a realistic system is in. It is only possible to measure the macrostate of the system.

A Statistical Definition of Temperature

- Two systems are in thermal contact with each other but thermally isolated from their surroundings:
- The i th system has energy E_i and can be in any one of $\Omega_i(E_i)$ microstates.
- The whole system (two systems combined) can be in any one of $\Omega_1(E_1)\Omega_2(E_2)$ microstates.
- The systems are able to exchange energy with each other and we will assume that they have been joined together for a sufficiently long time that they have come into thermal equilibrium (E_1 and E_2 have fixed values.)

A Statistical Definition of Temperature

- Crucial insight: *A system will appear to choose a macroscopic configuration that maximises the number of microstates.*
- Based on the following assumptions:
 - Each one of the possible microstates is equally likely.
 - The system's internal dynamics are such that the microstates of the system are continually changing.
 - Eventually, the system will explore all possible microstates and spend an equal time in each of them
- The system will most likely be found in a configuration that is represented by the most number of microstates.

“For a large system, ‘most likely’ becomes *overwhelmingly likely*.. to the level where it becomes an utterly reliable prediction on whose basis you can design an aircraft engine and trust your life to it! ”

A Statistical Definition of Temperature

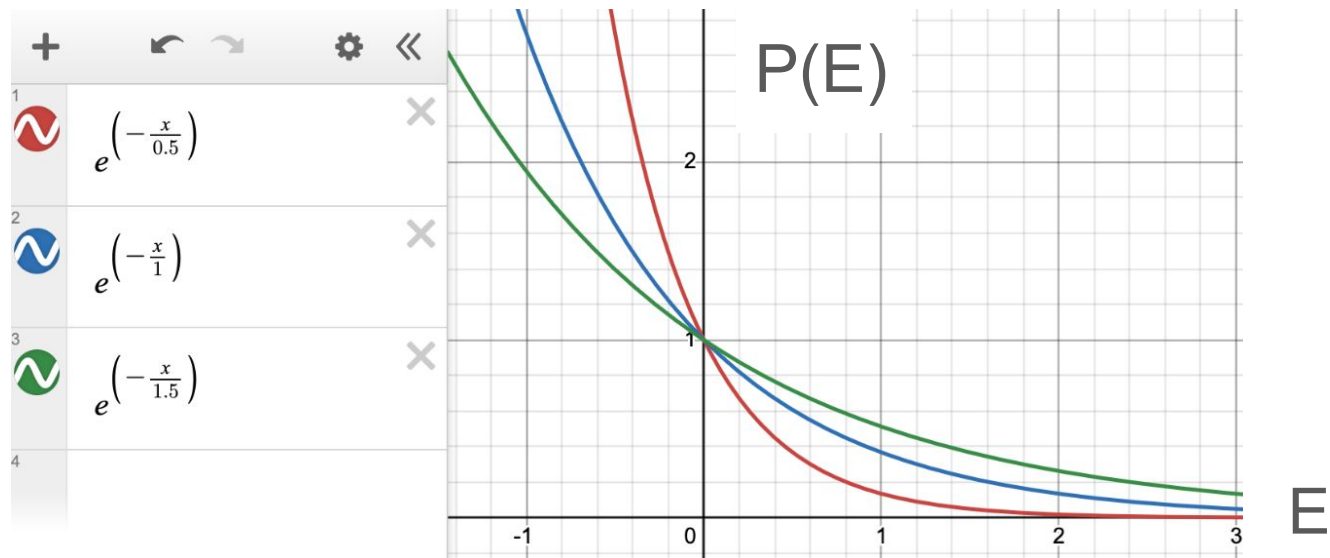
- For our two connected systems, the most probable division of energy between the two systems is the one that maximises $\Omega_1(E_1)\Omega_2(E_2)$.
- We can maximise this expression with respect to E_1

A Statistical Definition of Temperature

Boltzmann Distribution

- **Boltzmann distribution** (also called Gibbs distribution) is a probability distribution that gives the probability that a system in thermal equilibrium will be in a certain state as a function of that state's energy and the temperature of the system.

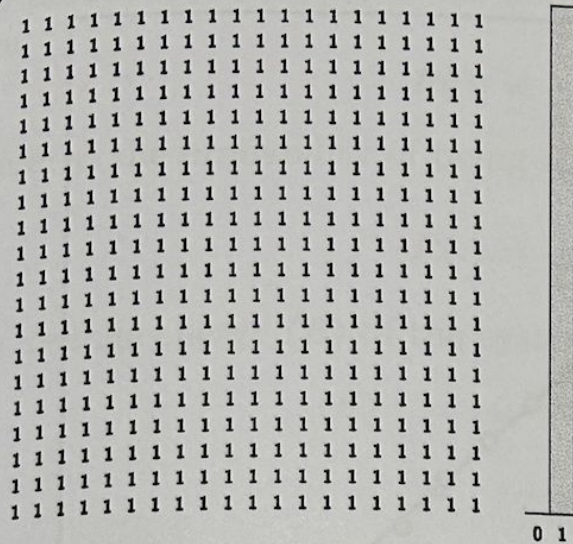
$$P(E) \propto \exp\left(\frac{-E}{k_B T}\right)$$



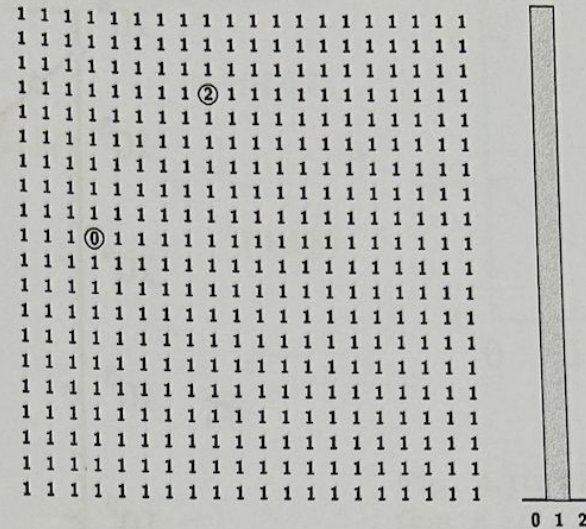
Boltzmann Distribution

Concepts in Thermal Physics S. J. Blundell and K. M. Blundell

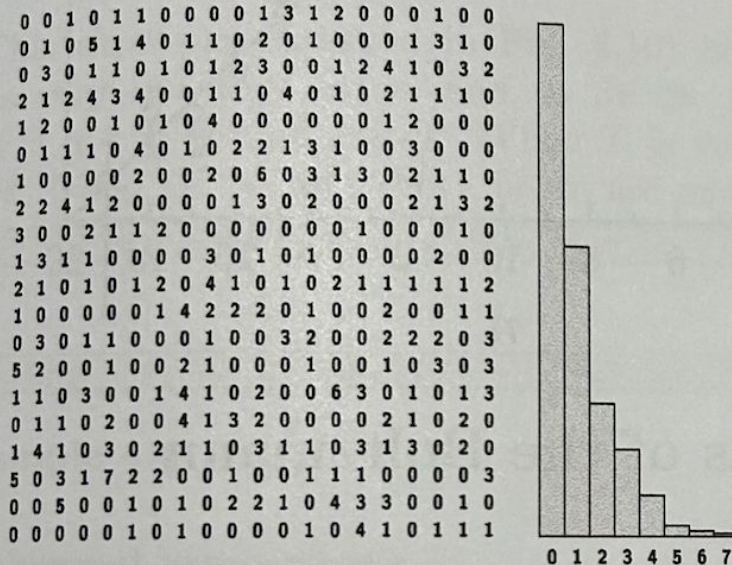
(a)



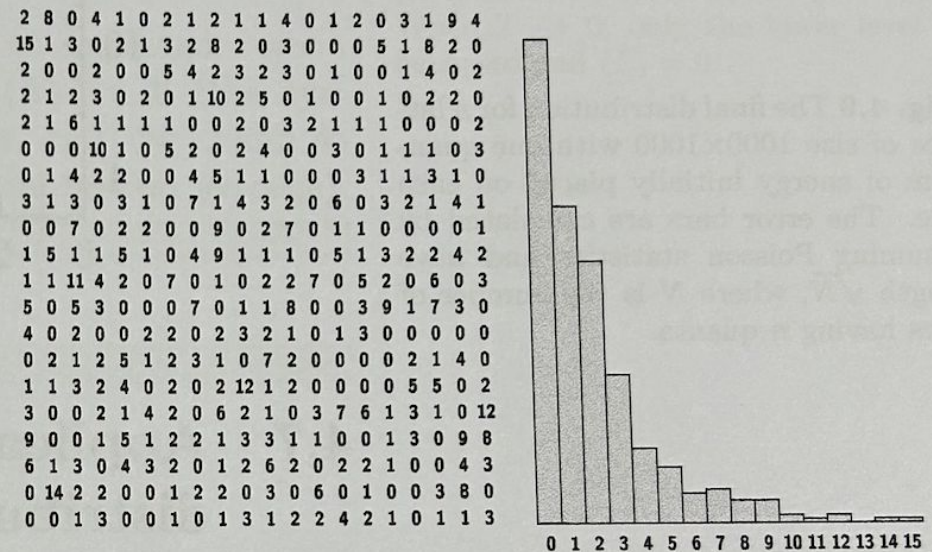
(b)



(c)



(d)



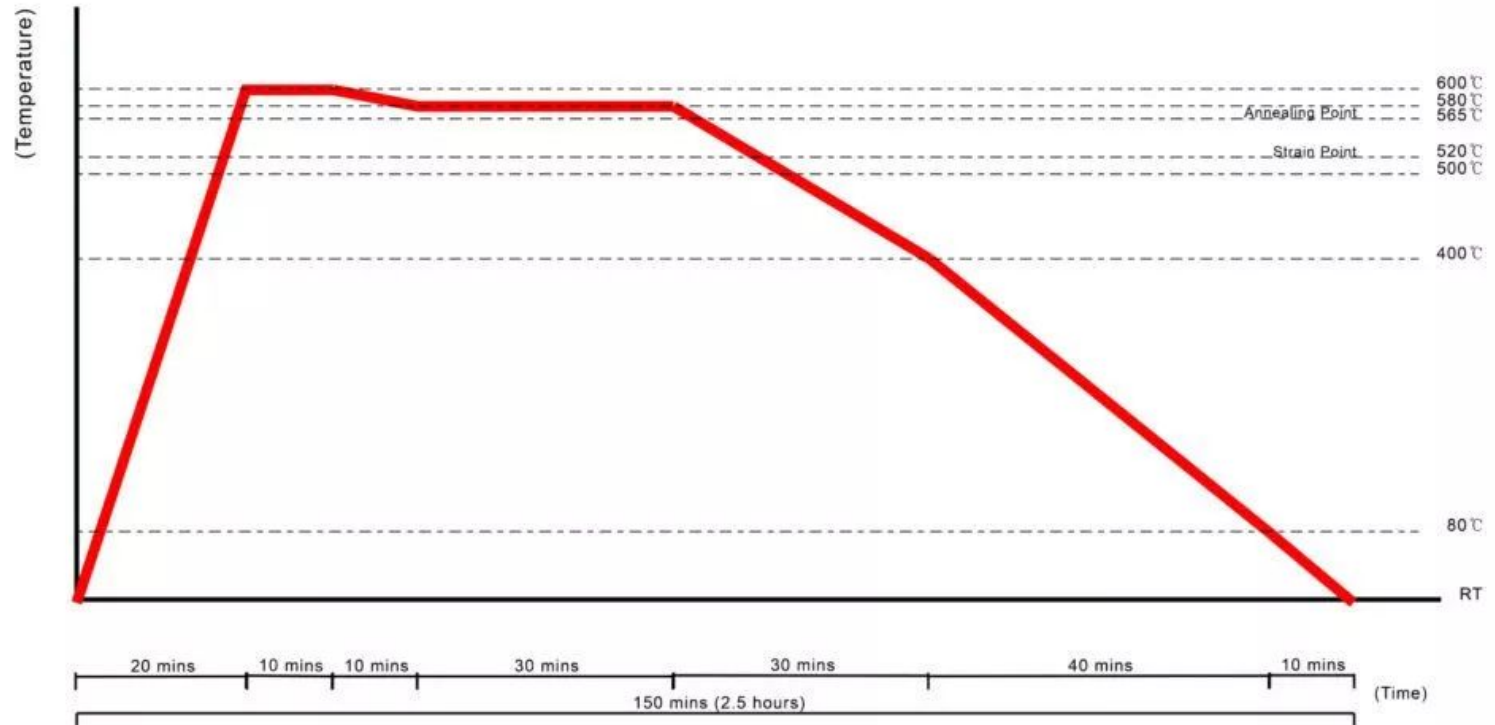
Simulated Annealing

Simulated Annealing

Kirkpatrick et al., Optimization
by simulated annealing, 1983.

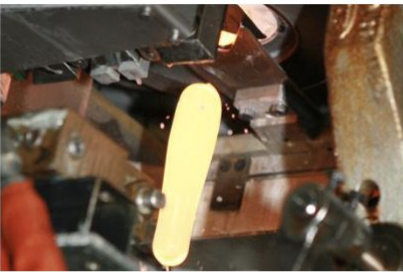
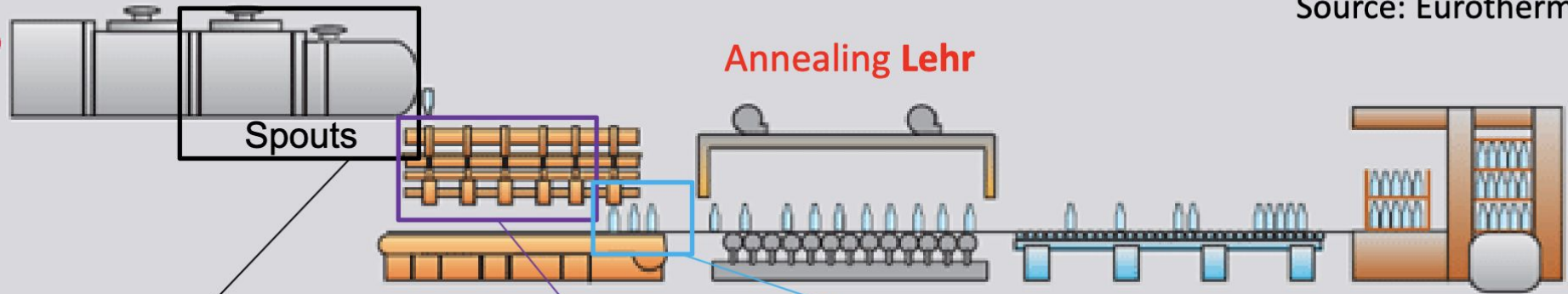
- The algorithm is inspired by the process of annealing.
- Atoms in a material form a crystal lattice structure. If the material is heated, the atoms are in a rapid state of motion. The motion randomises their positions and prevents them coming to rest in an orderly arrangement.
- As the material cools down, the atoms slow down, and if the cooling is slow enough, i.e., **annealing**, the atoms reconfigure into a minimum-energy state.
- If the material is cooled rapidly, i.e., **quenched**, the metal re-crystallises into a different higher-energy state (called an amorphous state).

Annealing Schedule



CONTAINER GLASS

Source: Eurotherm



Source: BDF Industries

Molding



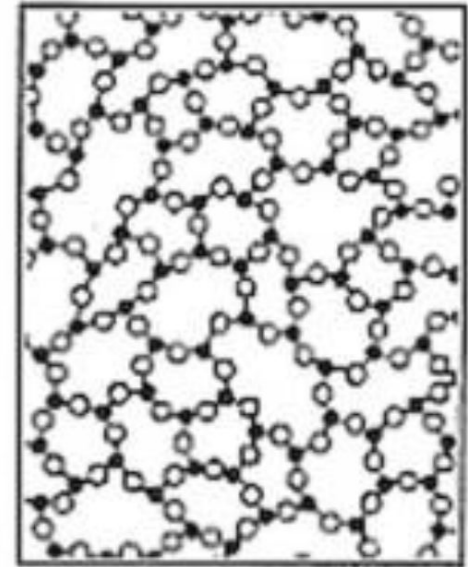
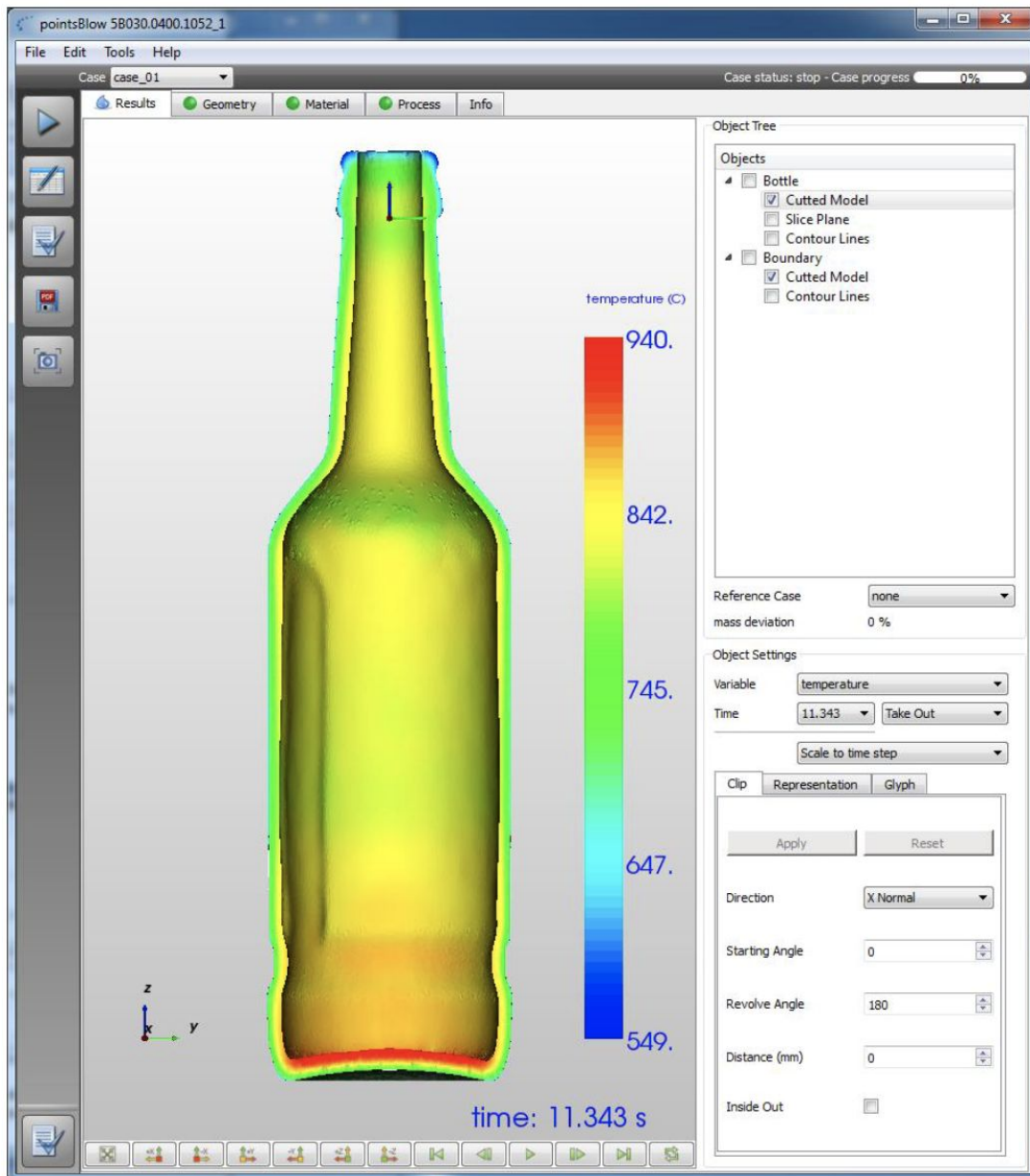
"Solid" glass

Conveying Belt

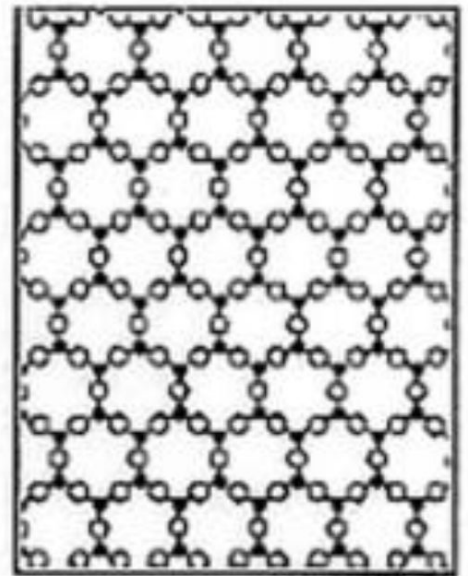


Further cooling during conveying

IMI-NFG Course on Processing of Glass - Lecture 9: Annealing and Tempering
mathieu.hubert@celsian.nl



Disordered Structure

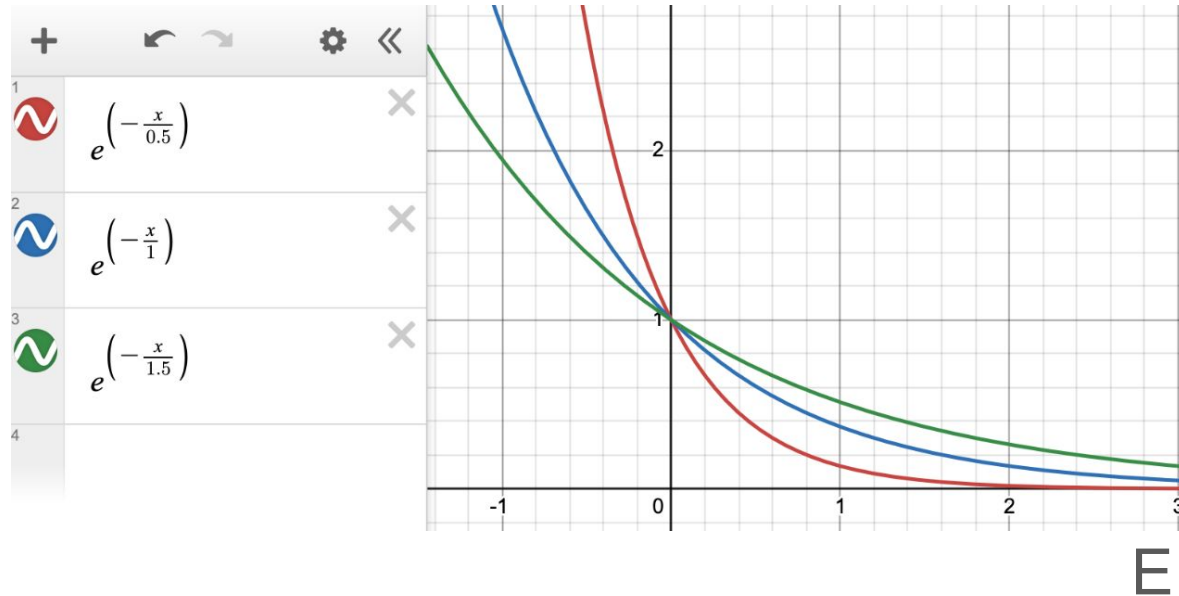


Ordered Structure

Simulated Annealing

- From statistical mechanics, the Boltzmann distribution describes the probability of a system occupying a given energy state:

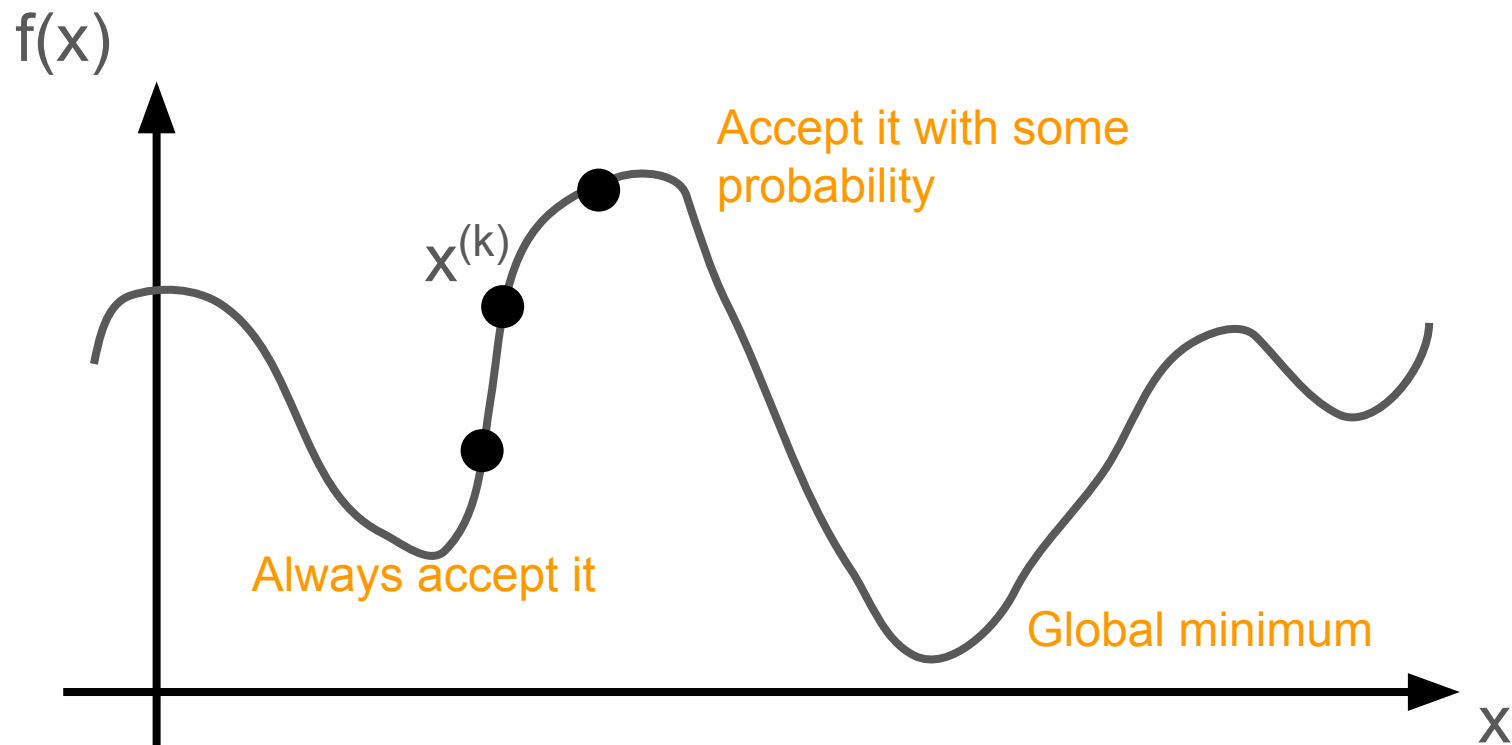
$$P(E) \propto \exp\left(\frac{-E}{k_B T}\right)$$



- This equation shows that as the temperature decreases, the probability of occupying a higher-energy state decreases, but is not zero.
- Therefore an atom could jump to a higher-energy state with some small probability, unlike in classical mechanics.

Simulated Annealing

- This probabilistic property imparts an exploratory nature to the optimisation algorithm.



Simulated Annealing

- Objective function value = Energy level
- Temperature is a parameter controlled by the optimiser, which begins high and slowly “cooled” to drive convergence.
- At the k th iteration, the design variables are given by $x^{(k)}$, and the objective function value (or energy) is given by $f(x^{(k)})$.
- A new state x_{new} is selected. If the energy level decreases, the new state is accepted. If the energy level increases, the new state might still be accepted with probability

$$\exp \left(\frac{-(f(x_{\text{new}}) - f(x^{(k)}))}{T} \right)$$

otherwise the state remains unchanged.

Simulated Annealing

- A new state x_{new} is usually selected at random in the **neighbourhood** of x .
- A neighbouring state is usually related to the current state compared to picking a pure random state from the entire set. In defining the neighbourhood structure, we might wish to define transition probabilities so that all neighbours are not equally likely. This type of structure is common in **Markov chain** problems.

Metropolis Algorithm

$M(RT)^2$ (1953) Equation of state calculations by fast computing machines.

THE JOURNAL OF CHEMICAL PHYSICS

VOLUME 21, NUMBER 6

JUNE, 1953

Equation of State Calculations by Fast Computing Machines

NICHOLAS METROPOLIS, ARIANNA W. ROSENBLUTH, MARSHALL N. ROSENBLUTH, AND AUGUSTA H. TELLER,
Los Alamos Scientific Laboratory, Los Alamos, New Mexico

AND

EDWARD TELLER,* *Department of Physics, University of Chicago, Chicago, Illinois*
(Received March 6, 1953)

A general method, suitable for fast computing machines, for investigating such properties as equations of state for substances consisting of interacting individual molecules is described. The method consists of a modified Monte Carlo integration over configuration space. Results for the two-dimensional rigid-sphere system have been obtained on the Los Alamos MANIAC and are presented here. These results are compared to the free volume equation of state and to a four-term virial coefficient expansion.

- Sampling algorithm
- Top 10 Algorithm of the 20th century

Metropolis Algorithm

M(RT)^2 (1953) Equation of state calculations by fast computing machines.

- Suppose we are in state i and we generate a new state j by making some change to state i .
- We choose the particular change uniformly at random from a set of possible changes - a **move set**.
- We then either **accept** or **reject** the new state with **acceptance probability** P_a given by:

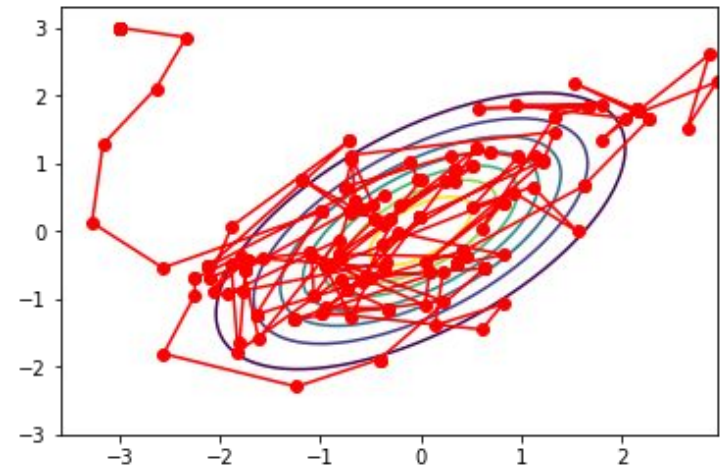
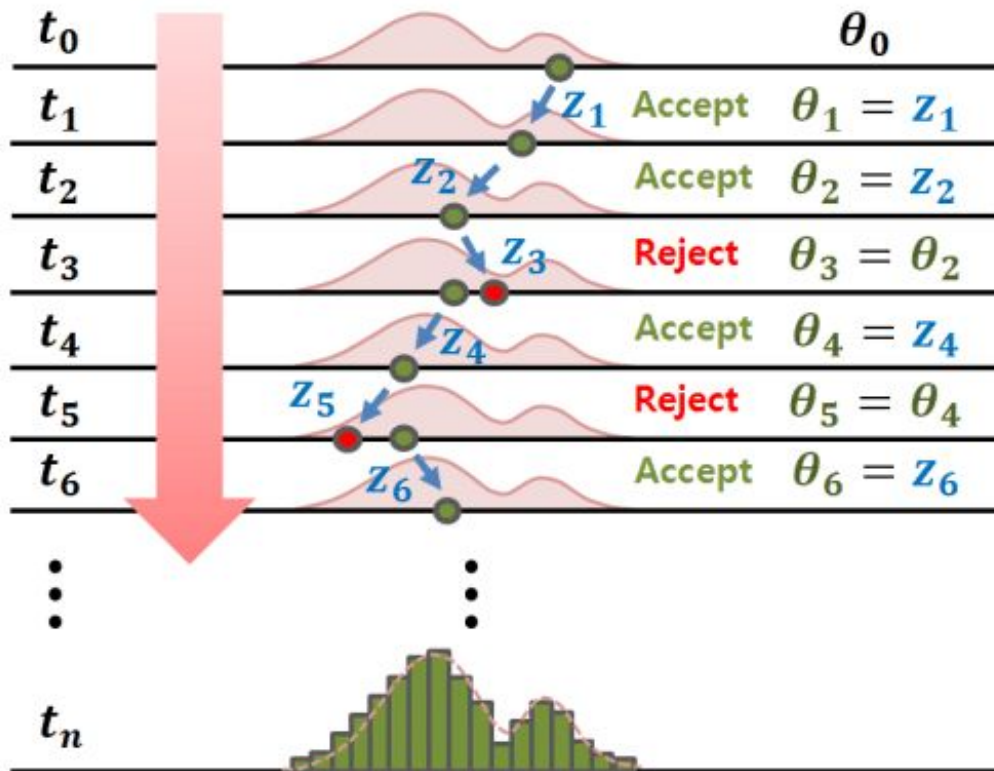
$$P_a = \begin{cases} 1 & \text{if } E_j \leq E_i \\ e^{-\beta(E_j - E_i)} & \text{if } E_j > E_i \end{cases}$$

- If the move is rejected then we do nothing, if move accepted we change the system to the new state

Metropolis Algorithm

$M(RT)^2$ (1953) Equation of state calculations by fast computing machines.

$$P_a = \min \left(1, e^{-\beta(E_j - E_i)} \right)$$



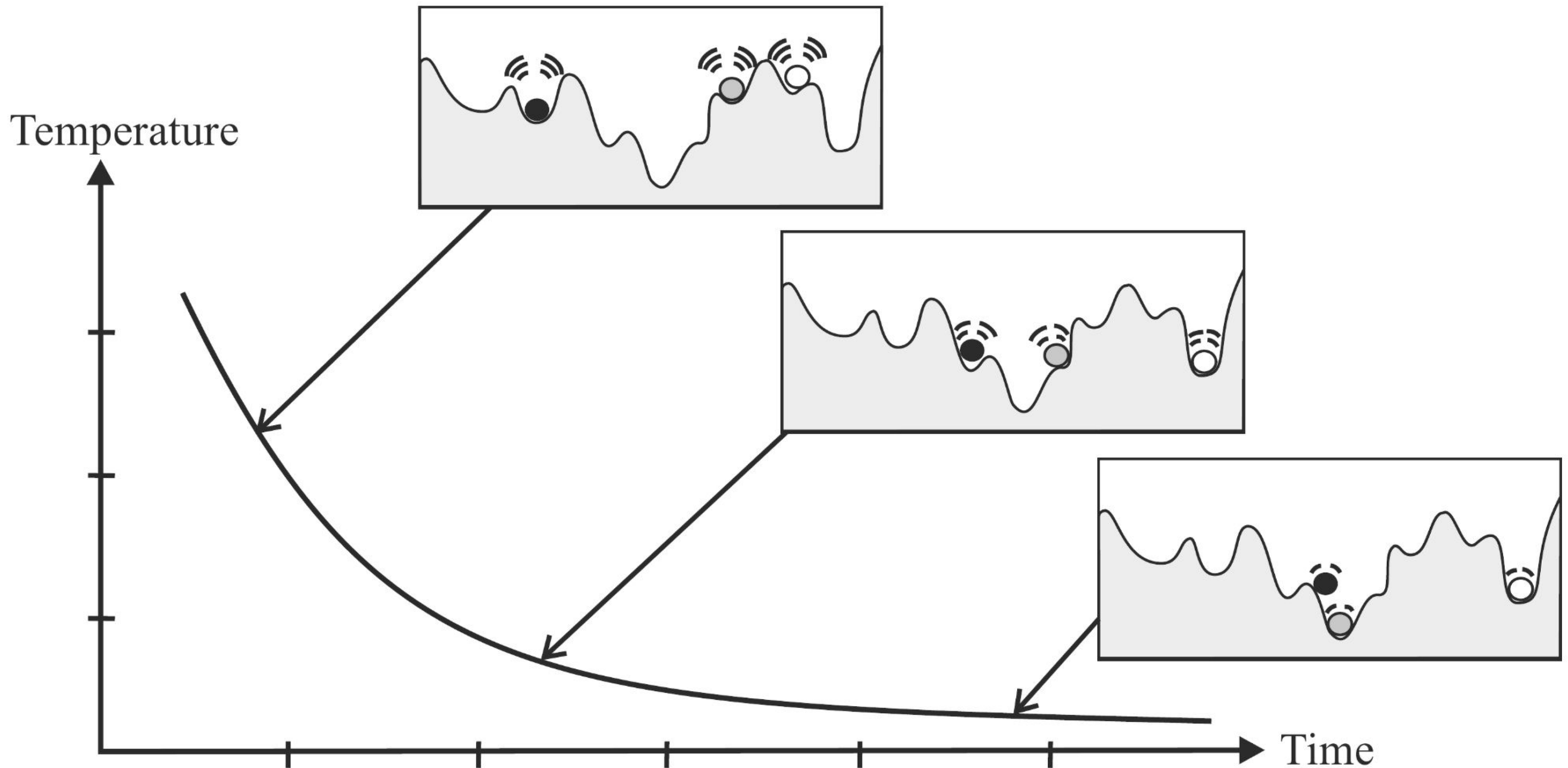
Metropolis Algorithm

$M(RT)^2$ (1953) Equation of state calculations by fast computing machines.

- The complete Markov chain Monte Carlo (MCMC) simulation involves the following steps:
 1. Choose a random starting state.
 2. Choose a move uniformly at random from an allowed set of moves.
 3. Calculate the acceptance probability P_a
 4. With probability P_a accept the move: the state of the system changes to the new state; otherwise reject it: system stays in its current state for one more step of the calculation.
 5. Measure the value of interest in the current state.
 6. Go to step 2.

Cooling

- The temperature level provides control on the level of expected exploration.



Cooling

- **Cooling schedule, or annealing schedule** - process for decreasing the temperature throughout the optimisation.
1. A common approach is an exponential decrease where T_0 is the initial temperature, α is the cooling rate, and k is the iteration number. The cooling rate α is close to 1, such as 0.8~0.99:

$$T = T_0 \alpha^k$$

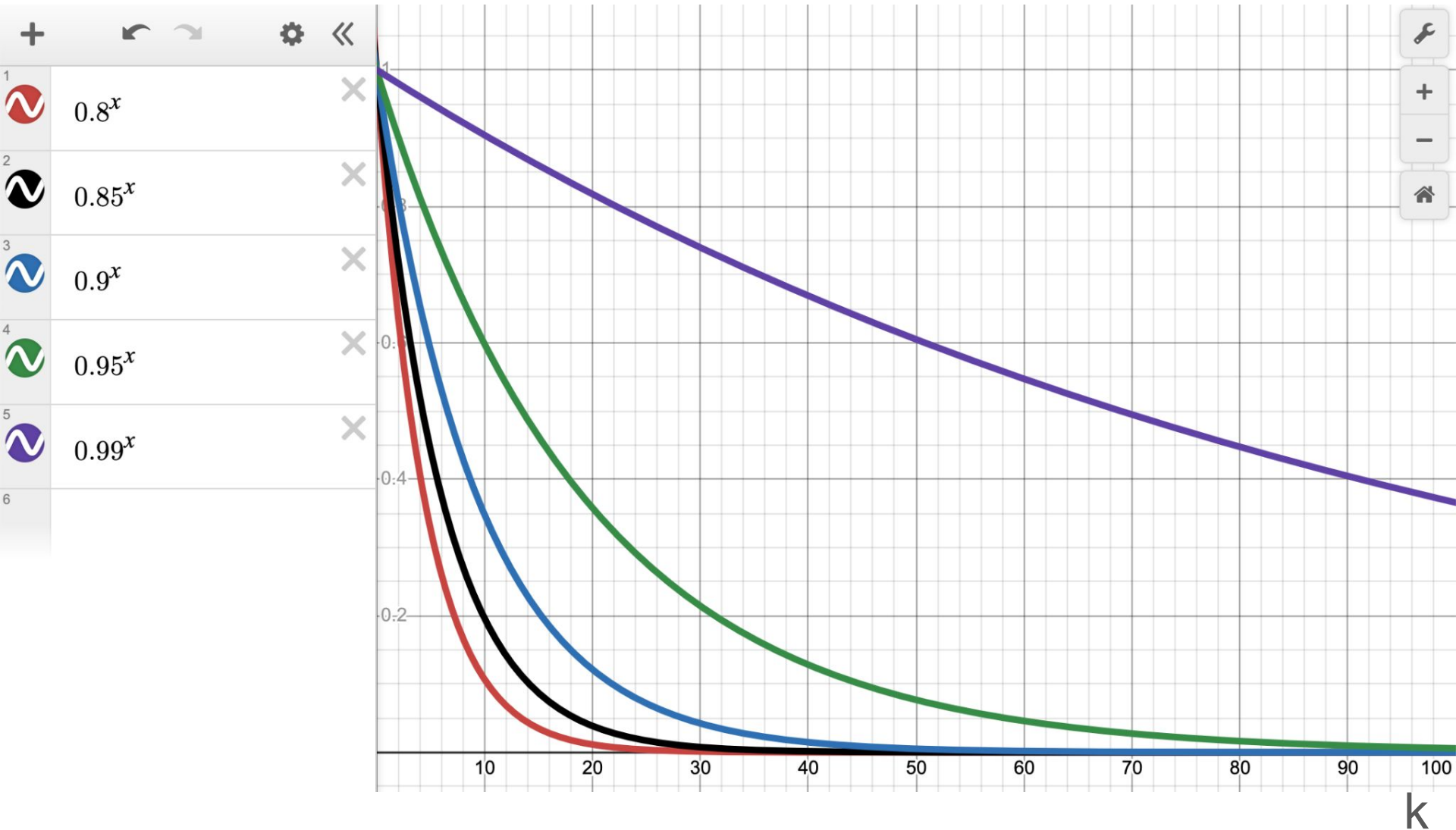
2. Another simple approach is the equation below where the exponent β is usually in the range of 1~4. A higher exponent corresponds to spending more time at low temperatures:

$$T = T_0 \left(1 - \frac{k}{k_{\max}} \right)^\beta$$

3. The temperature is kept constant for a fixed number of iterations, or successful moves, before decreasing.

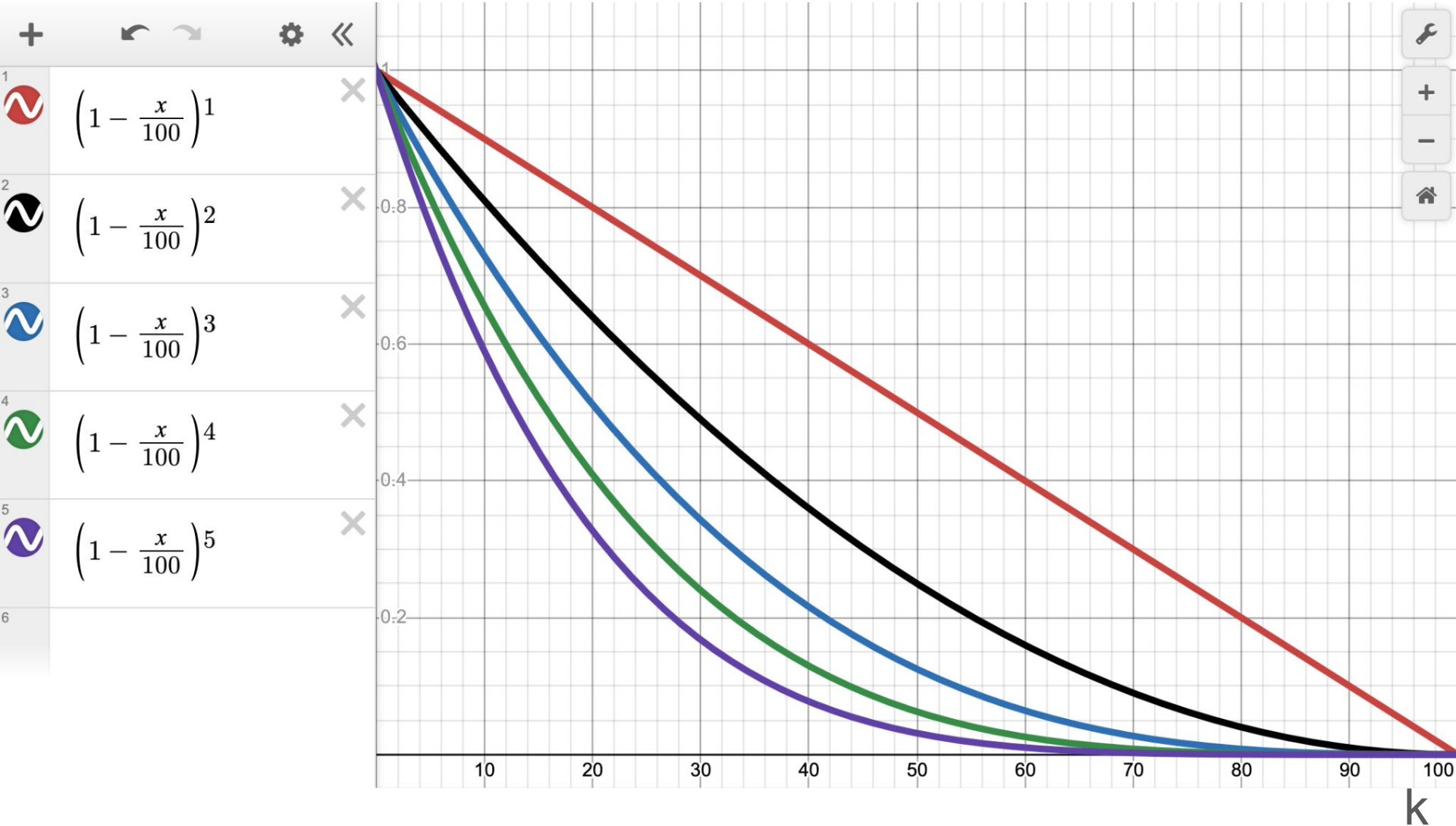
Cooling

$$T = T_0 \alpha^k$$



Cooling

$$T = T_0 \left(1 - \frac{k}{k_{\max}} \right)^{\beta}$$



Cooling

- Cooling should occur slowly to improve the ability to recover from a local optimum, imitating the annealing process instead of the quenching process.
- The cooling schedule can substantially impact the algorithm's performance, and some experimentation is usually required to select an appropriate schedule for the problem at hand.

Simulated Annealing

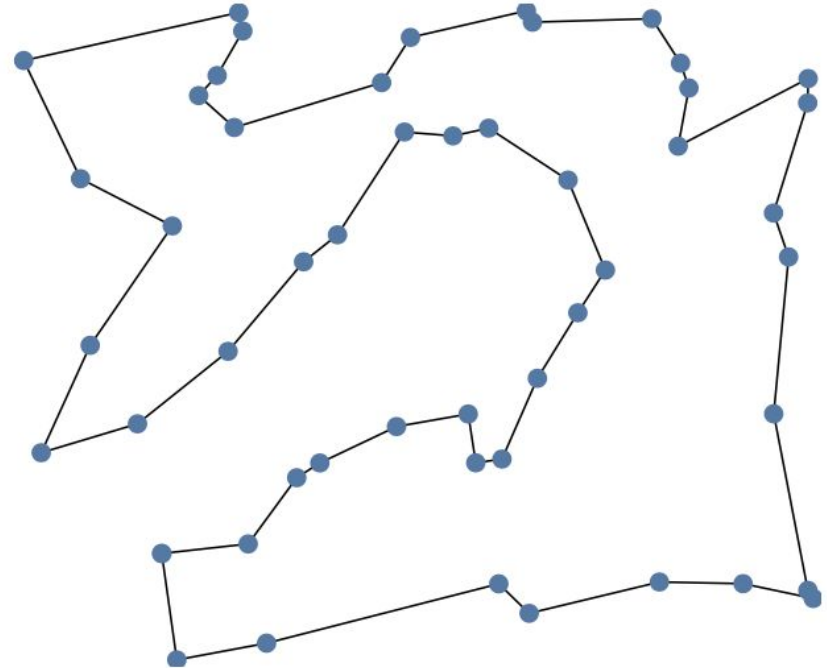
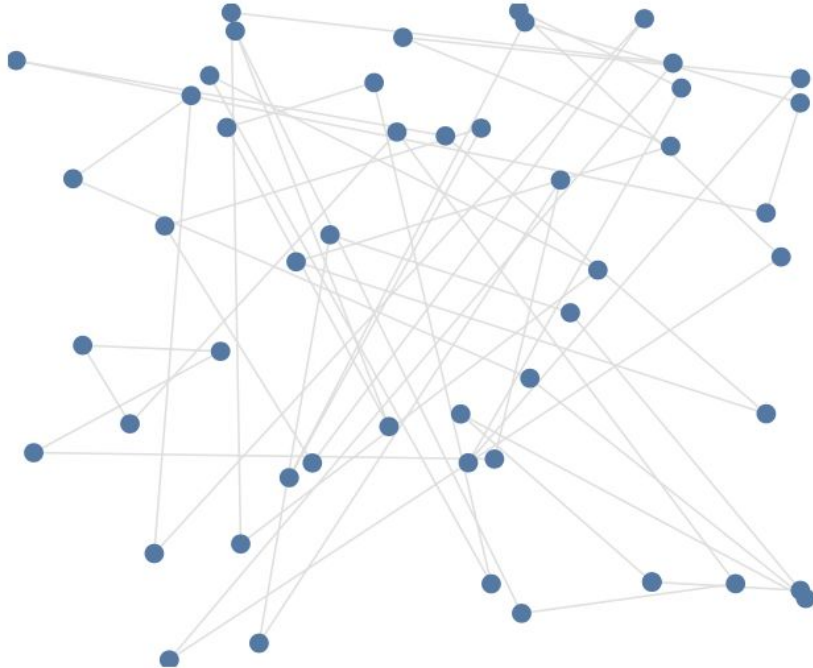
- Constraints can be handled in simulated annealing without resorting to penalties by rejecting any infeasible states.
- Can be applied to continuous multimodal problems as well. Motivation is similar: Initial high temperature permits escape from local minima. By slowly cooling, the initial exploration gives way to exploitation. Only change is in the neighbour function. A typical approach is to generate a random direction and choose a step size proportional to the temperature. Thus, smaller, more conservative steps are taken as the algorithm progresses.
- Purely random step directions are not particularly efficient for many continuous problems. One variation adopts concepts from the Nelder-Mead algorithm to improve efficiency.

Press et al., Numerical Recipes in C: The Art of Scientific Computing, 1992.

Applications

Application 1 - Travelling Salesperson

- The objective is the total Euclidean distance of a path that traverses all points and returns to the starting point.
- The design variables are a sequence of integers corresponding to the order in which the salesperson traverses the points.



Application 1 - Travelling Salesperson

Joe Chen / teaching

main ▾

teaching / enel445 / Simulate Annealing.ipynb

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 **Simulate Annealing.ipynb**  95.09 KiB





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Travelling Salesperson Problem solved by simulated annealing.

Date Created: 29 Mar 2024

Last Modified: 11 May 2024

Humans Responsible: JC

In [1]:

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(41)
```

In [2]:

```
N = 30      # Number of cities
Tmax = 10.0  # Max temp
Tmin = 1e-3  # Min temp
tau = 1e3    # How fast to cool
t_max = 10000 # Iteration budget
```

In [3]:

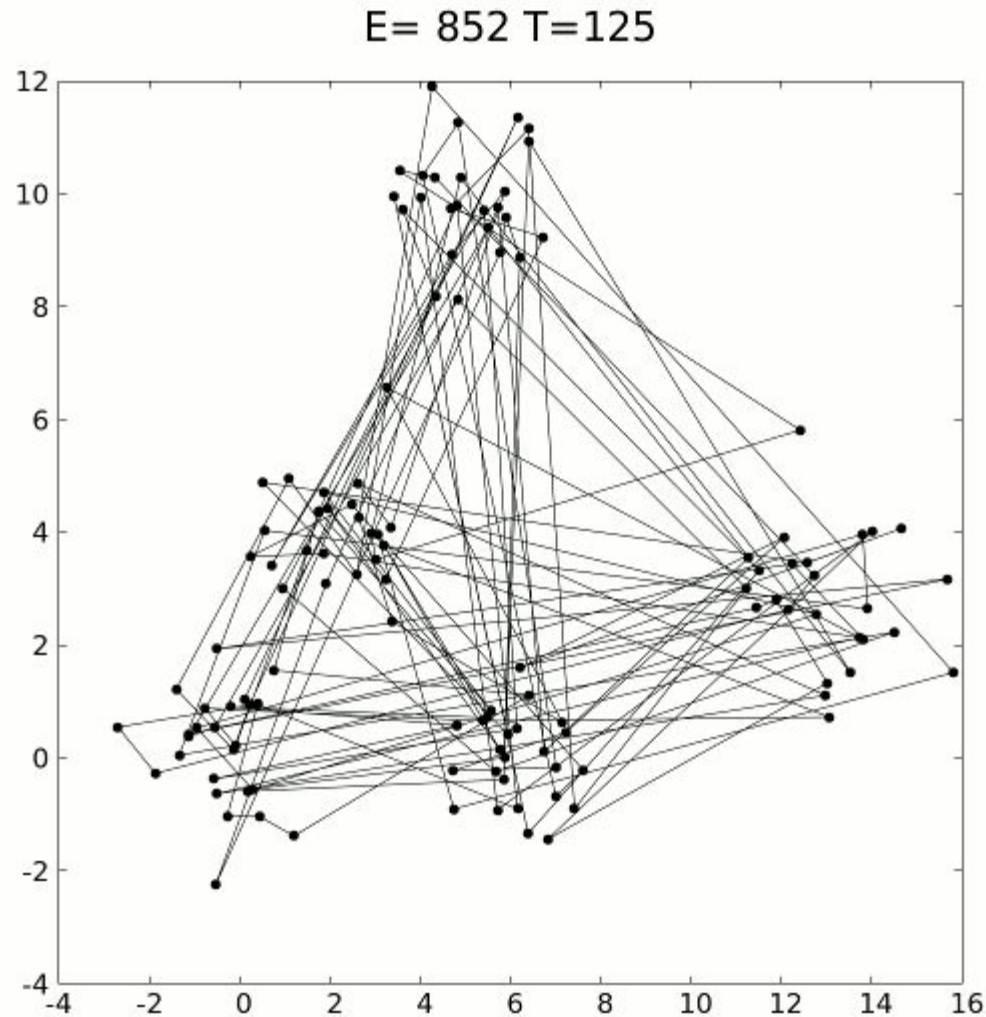
```
def get_total_distance(r):
    s = 0.0
    for i in range(N-1):
        s += np.sqrt(np.sum((r[i+1,:] - r[i,:])**2))

    # Add on the distance from the last city back to the first, since the salesman returns to their own city at the end
    s += np.sqrt(np.sum((r[N-1,:] - r[0,:])**2))
```

https://eng-git.canterbury.ac.nz/jch425/teaching/-/tree/main/enel445?ref_type=heads

Application 1 - Travelling Salesperson

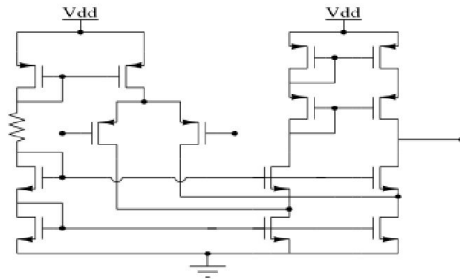
- The final path might not be the global optimum - these finite-time methods are only approximations of the full combinatorial search.
- But the methodology is effective and fast for this problem in finding a near-optimal solution.
- Stochastic method - fix seed for reproducibility



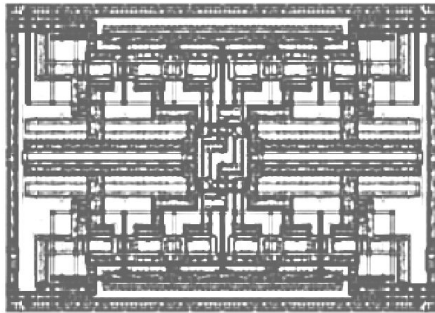
Application 2 - Physical Design of Computers

- Original motivation of Kirkpatrick et al.

Circuit Level - Example of a Folded Cascode Amplifier



Layout Level - Example of an Amplifier Layout



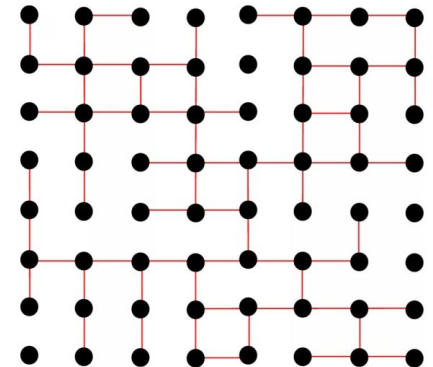
<https://www.intechopen.com/chapters/38549>

Simulated Annealing Technique for Routing in a Rectangular Mesh Network

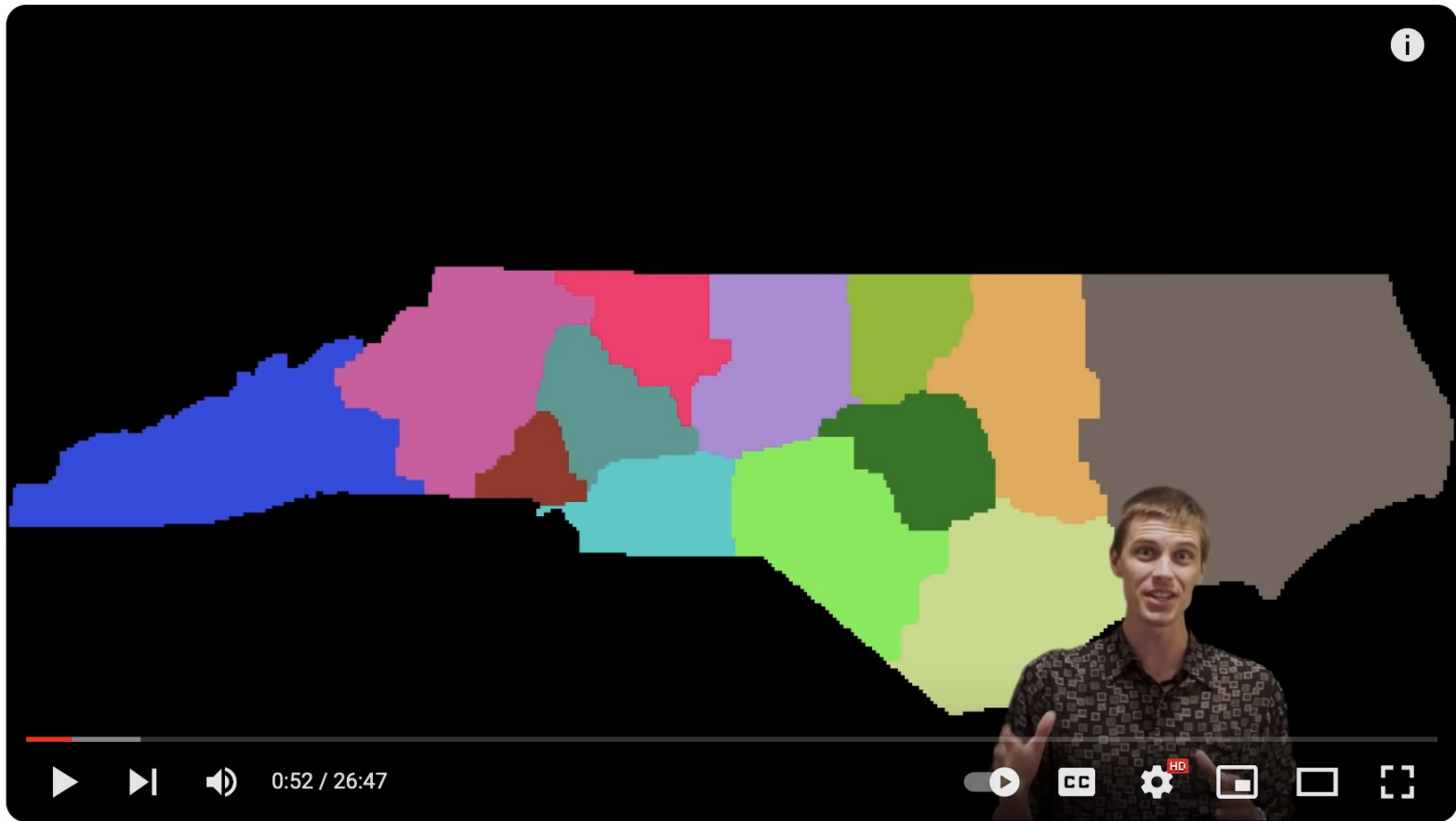
<https://www.hindawi.com/journals/mse/2014/127359/>

Physical Design of Computers

The physical design of electronic systems and the methods and simplifications employed to automate this process have been reviewed (11, 12). We first provide some background and definitions related to applications of the simulated annealing framework to specific problems that arise in optimal design of computer systems and subsystems. Physical design follows logical design. After the detailed specification of the logic of a system is complete, it is necessary to specify the precise physical realization of the system in a particular technology.



Application 3 - Ordering an Election



Algorithmic Redistricting: Elections made-to-order



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411,588 views Jun 28, 2021 #Gerrymandering #Math #Politics

<https://www.youtube.com/watch?v=Lq-Y7crQo44&t=1394s>

Application 4 - Untangling Earphones



Algorithmic Redistricting: Elections made-to-order



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<https://www.youtube.com/watch?v=Lq-Y7crQo44&t=1394s>

Exercises

- Look up the terms that are new to you in this lecture and understand them.
- Think of a problem and apply the simulated annealing algorithm to it.
- Try and combine simulated annealing with Nelder-Mead, Particle Swarm Optimisation or any other algorithm(s) you've learnt so far.
How would your hybrid algorithm proceed?

Thanks!